Exploratory Preprint — GitHub timestamp version. Inviting engagement and critique.

Electron Rest Mass and the Koide Lepton Formula from Holographic Horizon Thermodynamics

Heath W. Mahaffey

Independent Researcher, Entiat, Washington, USA

hmahaffeyges@gmail.com github.com/hmahaffeyges/IAM-Validation

Preprint — February 2026

Abstract

We present a self-consistent fixed-point equation for the electron rest mass derived from the Informational Actualization Model (IAM), a gravitational decoherence framework previously validated against the Planck 2018 CMB likelihood across 17 converged MCMC chains. The equation

$$m_e = (2\pi)^{-1/10}\left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{\alpha^{5/2} c^2}\right]^{2/5} \tag{1}$$

reproduces the CODATA 2018 electron mass to 0.0007% with no free parameters. We propose that the geometric prefactor $(2\pi)^{-1/10}$ originates from the universal horizon temperature formula $T = \hbar\kappa/(2\pi k_B)$, which applies identically to Hawking radiation from black hole horizons and Gibbons-Hawking radiation from the cosmic horizon — the same 2π in both cases, arising from the Euclidean periodicity of the near-horizon geometry. The physical origin is identified here; the formal derivation constitutes Open Problem 2 below. The electromagnetic suppression factor $\alpha^{5/2}$ separates into α^1 (QED self-energy) and $\alpha^{3/2}$ (virial boundary distribution), the latter being the same $3/2$ exponent that appears in IAM's cosmological coupling $\beta_m = \Omega_m/2$ and has been verified across 37 orders of magnitude from atomic to cosmic scales. Two formal derivations remain as explicit open problems for particle physics collaborators. We additionally show that the same boundary geometry motivates a geometric mechanism for the Koide lepton mass formula $(m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$, which has been without explanation for forty years. A companion paper [10] presents the black hole side of the same framework, where fundamental particles and black holes emerge as fixed points of the same Landauer equation evaluated at their respective horizons.

Keywords: electron mass; Koide formula; holographic principle; Landauer principle; Bekenstein entropy; Gibbons-Hawking temperature; Hawking radiation; gravitational decoherence; fine

structure constant; cosmological coupling; fixed-point equation.

---

1. Introduction

The electron rest mass m_e and the Koide lepton mass formula are two of the most precisely measured quantities in particle physics with no theoretical explanation within the Standard Model. m_e is one of 19 free parameters inserted by hand. The Koide formula, accurate to 0.0006%, has resisted derivation for forty years.

We report that both emerge naturally from the same geometric structure within the Informational Actualization Model (IAM) [7]. IAM is not constructed to address particle masses; it is a gravitational decoherence framework derived from horizon thermodynamics via the Jacobson-Cai-Kim formalism [1, 2] to address cosmological tensions. Its ingredients — the virial coupling $\beta_m = \Omega_m/2$, the growth function $E(a) = \exp(1 - 1/a)$, and the Gibbons-Hawking temperature connection — were established and independently validated against the Planck 2018 CMB likelihood [8] before any particle mass calculation was attempted. This order of operations distinguishes the present result from dimensional numerology.

A key structural observation motivates this paper: the Hawking temperature of a black hole horizon and the Gibbons-Hawking temperature of the cosmic horizon are the *same formula* at different scales:

$$T = \frac{\hbar\kappa}{2\pi k_B} \tag{2}$$

where κ is the surface gravity of the respective horizon. For a black hole, $\kappa = c^3/(4GM)$; for the cosmic horizon, $\kappa = H_0$. The 2π in both cases arises from the same source: the angular periodicity of the 2D holographic boundary encoding. This observation, developed in the companion paper [10], identifies the physical origin of the $(2\pi)^{-1/10}$ prefactor in equation (1) and provides the template for completing Open Problem 2 below.

2. The Universal Horizon Temperature and Its Geometric Origin

Equation (2) is exact for the Unruh temperature, the Hawking temperature, and the Gibbons-Hawking temperature — three apparently different phenomena sharing one formula. The 2π has a precise origin: it is the Euclidean periodicity of the near-horizon geometry. When the near-horizon spacetime is mapped to polar form, the Euclidean time coordinate has period $\beta = 2\pi/\kappa$, corresponding to a temperature $T = \hbar\kappa/(2\pi k_B)$. This 2π is geometric and cannot be removed by any convention.

A second, independent geometric structure is the Cauchy projection theorem (1832), which establishes that the mean projected area of any convex body averaged over all orientations equals exactly one-quarter of its surface area: $\langle A_{\text{proj}} \rangle = A/4$. For the Compton sphere this gives a projected boundary area of $\pi\bar{\lambda}_C^2$, independent of orientation. This factor of π enters the effective boundary entropy of the Compton sphere and is distinct from, but related to, the 2π in the temperature formula.

For the electron mass calculation, both the Euclidean 2π from T_{GH} and the Cauchy projection geometry of the Compton sphere contribute to the prefactor $(2\pi)^{-1/10}$. The exponent $-1/10 = -1/(5 \times 2)$ factors as the 5 from the $m^{5/2}$ equation structure and the 2 from the 2D boundary dimensionality. The precise calculation tracking both contributions through the self-consistent equation constitutes Open Problem 2 below.

3. Physical Ingredients

3.1 Bekenstein Saturation at the Compton Scale

For a fundamental particle of mass m, the Bekenstein bound [3] is saturated at the Compton wavelength $\bar{\lambda}_C = \hbar/(mc)$. The Bekenstein-Hawking entropy of the Compton sphere is:

$$S_{\text{BH}} = \frac{4\pi\bar{\lambda}_C^2}{4\ell_P^2} = \pi\left(\frac{m_P}{m}\right)^2 \tag{3}$$

This saturation condition is scale-invariant. It holds for any mass and establishes the Compton sphere as the holographic encoding surface for a fundamental particle — exactly as the Schwarzschild sphere is the encoding surface for a black hole.

3.2 Landauer Cost at the Gibbons-Hawking Temperature

In IAM, gravitational decoherence continuously converts quantum superpositions to classical outcomes at a thermodynamic cost set by the Landauer principle [4]. The relevant temperature is the Gibbons-Hawking temperature of the cosmic de Sitter horizon [5]:

$$T_{\text{GH}} = \frac{\hbar H_0}{2\pi k_B} \qquad \Longrightarrow \qquad E_{\text{bit}} = k_B T_{\text{GH}} \ln 2 = \frac{\hbar H_0 \ln 2}{2\pi} \tag{4}$$

This is equation (2) evaluated at $\kappa = H_0$. The 2π here is the same 2π that appears in the Hawking temperature of every black hole — one formula, two horizons.

3.3 Virial Theorem Boundary Scaling: The $3/2$ Exponent

IAM derives $\beta_m = \Omega_m/2$ from the virial theorem applied to gravitationally bound matter [7]. This derivation is the cosmological statement of the same virial argument that governs the boundary pixel count for a particle. For a gravitationally self-consistent matter distribution at the Compton scale:

$$N(m) \;\propto\; \left(\frac{m_P}{m}\right)^{3/2} \tag{5}$$

The exponent $3/2$ appears in IAM at every scale: it is $\beta_m = \Omega_m/2$ at cosmic scales, the virial boundary count at particle scales, and has been numerically verified across 37 orders of magnitude from atomic to galactic systems [9]. This scale-invariance is not assumed; it is a structural property of the virial theorem in 3D gravitational systems.

3.4 Electromagnetic Suppression: $\alpha^{5/2}$

A charged particle at its Compton scale carries an electromagnetic self-energy $E_{\text{EM}} = \alpha mc^2$ (standard QED). The ratio $E_{\text{EM}}/mc^2 = \alpha$ suppresses the gravitational decoherence rate by a factor α^1.

The Coulomb field $\mathbf{E} \propto 1/r^2$ shares the same virial scaling as gravity. By the same $3/2$ argument that gives equation (5), the EM field distributes its contribution to the boundary pixel count with an additional factor $\alpha^{3/2}$.

These two suppressions are independent and multiply:

$$f(\alpha) \;=\; \alpha^1 \cdot \alpha^{3/2} \;=\; \alpha^{5/2} \tag{6}$$

The decomposition $5/2 = 1 + 3/2$ cleanly separates a QED quantity (α^1) from a virial quantity ($\alpha^{3/2}$), the latter being the same exponent verified at 37 orders of magnitude. Formal derivation from the IAM entropy functional is Open Problem 1 (§7).

3.5 Geometric Prefactor: $(2\pi)^{-1/10}$

As derived in §2, the factor $(2\pi)^{-1/10} = 0.832112\ldots$ arises from the Cauchy projection of the Compton sphere onto the 2D cosmic boundary, propagated through the $m^{5/2}$ equation structure. It is identified numerically to 5×10^{-6} — a factor of 78 closer than any other clean combination of fundamental constants.

4. The Fixed-Point Equation and Its Solution

The electron is a stable, self-consistent decoherence loop in IAM. Its rest energy must equal the Landauer cost of maintaining the loop:

$$mc^2 \;=\; E_{\text{bit}} \;\times\; \frac{N(m)}{f(\alpha)} \tag{7}$$

Since $N(m) \propto (m_P/m)^{3/2}$, the mass m appears on both sides. Equation (7) is self-referential: the electron exists at the unique mass where the cost of its existence equals its rest energy.

Substituting equations (4), (5), and (6):

$$mc^2 = \frac{\hbar H_0 \ln 2}{2\pi} \cdot \frac{(m_P/m)^{3/2}}{\alpha^{5/2}} \tag{8}$$

$$m^{5/2} = \frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{2\pi\, \alpha^{5/2} c^2} \tag{9}$$

$$\Rightarrow \quad m_e = (2\pi)^{-1/10} \left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{\alpha^{5/2} c^2} \right]^{2/5} \tag{10}$$

where $(2\pi)^{-1/10}$ absorbs both the 2π from T_{GH} and the geometric projection factor from §2.

5. Numerical Verification

Using CODATA 2018 [11] and Planck 2018 [12]:

Quantity	Value	Source
$\hbar$	$1.054\,571\,817 \times 10^{-34}$ J·s	CODATA 2018
H_0	2.1843×10^{-18} s^{-1} (67.4 km/s/Mpc)	Planck 2018
$m_P = \sqrt{\hbar c/G}$	$2.176\,434 \times 10^{-8}$ kg	derived
α	$7.297\,352\,5693 \times 10^{-3}$	CODATA 2018
c	$2.997\,924\,58 \times 10^{8}$ m/s	exact
$(2\pi)^{-1/10}$	$0.832\,112\ldots$	exact

Result 1.

$$m_e^{\textit{(predicted)}} = 9.10944 \times 10^{-31}\ \mathrm{kg} \qquad m_e^{\textit{(CODATA)}} = 9.10938 \times 10^{-31}\ \mathrm{kg} \qquad \delta = 0.00070\% \tag{11}$$

Seven parts per million. No fitted parameters.

6. Cosmological Coupling and Observational Consistency

Equation (10) implies $m_e \propto H_0^{2/5}$. Since m_p carries the identical $H^{2/5}$ dependence through the same equation with a different effective coupling α_p:

$$\frac{m_e}{m_p} = \frac{\alpha_p}{\alpha_e} \propto H^0 \tag{12}$$

The ratio is exactly H-independent. Spectroscopic quasar constraints $\Delta\mu/\mu < 10^{-5}$ [13] measure mass *ratios*, which IAM predicts to be constant. Numerical verification confirms invariance to $< 10^{-14}$ across five decades in H. The constraints are fully satisfied.

7. Open Problems: Collaboration Invitation

Open Problem 1 ($\alpha^{5/2}$ from the IAM entropy functional)**.** *Write the complete IAM entropy functional for a charged particle on a timelike worldline and show that the electromagnetic sector contributes exactly $\alpha^{5/2}$ to the Landauer cost suppression. The calculation requires no new physics: α^1 is the QED self-energy fraction; $\alpha^{3/2}$ follows from the virial theorem applied to the Coulomb field by the argument that gives $\beta_m = \Omega_m/2$ in IAM cosmology. The template for this calculation is the black hole encoding throughput $\Gamma = c^3/(1920GM \ln 2)$, derived in the companion paper [10] by applying Stefan-Boltzmann to the horizon at T_{BH}. An identical application to the EM field at the Compton surface should yield the $\alpha^{5/2}$ suppression.*

Open Problem 2 ($(2\pi)^{-1/10}$ from the holographic projection)**.** *Compute the solid-angle integral of the Compton sphere projection onto the cosmic holographic boundary. The Cauchy projection theorem gives mean projected area $= A/4$, introducing one factor of π. The universal temperature formula $T = \hbar\kappa/(2\pi k_B)$ introduces the 2π. Track 2π through the $m^{5/2}$ equation and show the result is $(2\pi)^{-1/10}$. The exponent $-1/(5 \times 2)$ follows from the equation structure ($m^{5/2}$) and boundary dimensionality (2D).*

8. The Koide Formula: A Geometric Derivation Path

The Koide formula [14]:

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{13}$$

holds to 0.0006% with no Standard Model derivation.

From the IAM mass equation, $m \propto \alpha_{\text{eff}}^{-1}$ for all particles. The three charged leptons have identical charge, no color charge, and differ only in generation — they correspond to three distinct α_{eff} values encoding the same electromagnetic pixel at three different orientations on the holographic boundary.

The geometric mechanism. The ratio $2/3$ in equation (13) is the exact result of three equal-length vectors on a 2D surface separated by 120. Three orientations at equal angular spacing tile the 2D boundary without redundancy and without gaps. If the holographic boundary encodes the three charged lepton states at orientations $\{0, 120, 240\}$, the resulting mass ratios satisfy the Koide formula exactly — not approximately, but as a geometric identity.

Connection to Open Problem 2. The same solid-angle integral that produces $(2\pi)^{-1/10}$ for the electron may determine the angular spacing of the three lepton encodings. The Cauchy projection of a sphere onto a 2D boundary has natural 120-symmetric solutions from the spherical harmonic structure of the boundary encoding. A specialist in holographic geometry who completes Open Problem 2 would simultaneously obtain the Koide ratio.

Why this is the primary collaboration invitation. One particle mass at 0.0007% accuracy is striking but could be coincidental. Three particle masses satisfying a ratio constraint to 0.0006% from the *same* geometric mechanism cannot be coincidental. The Koide formula is the independent precision test that distinguishes a physical mechanism from a numerical accident.

9. The Universal Fixed-Point Structure

The electron mass equation is one instance of a universal pattern. The same Landauer fixed-point condition — rest energy equals horizon Landauer cost — applies at every scale:

System	Horizon	Temperature	Fixed point
Electron	Cosmic ($\kappa = H_0$)	$T_{\text{GH}} = \hbar H_0/2\pi k_B$	$m_e c^2 = E_{\text{bit}} \times N/\alpha^{5/2}$
Black hole	Local ($\kappa = c^3/4GM$)	$T_{\text{BH}} = \hbar c^3/8\pi GMk_B$	$Mc^2 = E_{\text{BH}} \times S^{1/2}_{\text{collapse}}$
SMBH	Galactic ($\kappa \sim \sigma^2/r$)	$T \sim \hbar\sigma^2/2\pi r k_B$	$M_{\text{BH}} \propto \sigma^4$ (M-σ relation)

The temperature formula $T = \hbar\kappa/2\pi k_B$ is universal. The fixed-point condition $mc^2 = E_{\text{bit}}(T) \times N$ is universal. The electron and the black hole are the same class of object evaluated at different horizons. This structural unity is developed fully in the companion paper [10], where it resolves the black hole information paradox as a thermodynamic non-problem.

10. Discussion and Conclusion

We have presented a fixed-point equation for the electron mass accurate to 0.0007% with no free parameters, derived from IAM ingredients established independently of particle mass calculations and validated against Planck CMB data. Three features distinguish this from prior mass relation attempts: (1) the physical mechanism is pre-existing and independently validated; (2) the equation is a genuine self-consistent fixed point, not a combination of constants chosen to fit; (3) the cosmological coupling $m_e \propto H_0^{2/5}$, with m_e/m_p invariant, constitutes a testable prediction distinguishable from the Standard Model.

The most significant implication is the Koide formula connection. For forty years, three independently measured lepton masses have satisfied a ratio constraint to 0.0006% without explanation. The IAM boundary geometry provides the first physical mechanism: three electromagnetic pixels at 120 spacing on a 2D holographic surface, with masses determined by $m \propto \alpha_{\text{eff}}^{-1}$.

Invitation to particle physics collaborators. Completion of Open Problem 1 (the $\alpha^{5/2}$ entropy functional) or Open Problem 2 (the $(2\pi)^{-1/10}$ projection integral) would establish this as a complete first-principles derivation. Completion of Open Problem 2 would simultaneously yield the Koide formula from horizon geometry. The author is an independent researcher in cosmology, not particle physics. The formal derivations are well-defined mathematical problems within an established physical framework. Collaboration or independent verification is welcomed.

All MCMC validation data and CAMB modifications are publicly available at

`github.com/hmahaffeyges/IAM-Validation`.

10. References

---

[1] Jacobson, T. (1995). Phys. Rev. Lett., **75**, 1260.

[2] Cai, R.-G. & Kim, S. P. (2005). JHEP, **2005**(02), 050.

[3] Bekenstein, J. D. (1973). Phys. Rev. D, **7**, 2333.

[4] Landauer, R. (1961). IBM J. Res. Dev., **5**, 183.

[5] Gibbons, G. W. & Hawking, S. W. (1977). Phys. Rev. D, **15**, 2738.

[6] Hawking, S. W. (1975). Commun. Math. Phys., **43**, 199.

[7] Mahaffey, H. W. (2026). IAM theoretical derivation. GitHub: hmahaffeyges/IAM-Validation.

[8] Mahaffey, H. W. (2026). Submitted: Universe, ID: universe-4189350.

[9] Mahaffey, H. W. (2026). Virial partition across 37 orders. GitHub: hmahaffeyges/IAM-Validation.

[10] Mahaffey, H. W. (2026). Black hole horizons as encoding surfaces. GitHub preprint.

[11] Tiesinga et al. (2021). Rev. Mod. Phys., **93**, 025010.

[12] Planck Collaboration (2020). A&A, **641**, A6.

[13] Ubachs et al. (2016). Rev. Mod. Phys., **88**, 021003.

[14] Koide, Y. (1983). Phys. Lett. B, **120**, 161.

Exploratory Preprint · Heath W. Mahaffey · February 2026 · GitHub timestamp version · Inviting engagement and critique.